



Technical Service Bulletin

Technical Service Bulletin: TSB130142

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CENSE™ CM530 Obsoleted and Replaced by Next Generation CENSE™ CM2330

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document announces that the CENSE™ CM530 module will be obsoleted in September 2013. The Next Generation CENSE™ CM2330 module will be superseding the existing CENSE™.

Calibrations for the Next Generation CENSE™ CM2330 module will be available through QuickServe™ Online and the INCAL™ CD. ECM Calibration Revision History for the CM2330 product will also be available through QSOL. A full list of old vs. new EI options and engine control module (ECM) codes for diesel and natural gas engine platforms can be found toward the end of this communication.

The Next Generation CENSE™ CM2330 module will **not** be supported by the INSITE™ CENSE™ electronic service tool. Field personnel **must** use base INSITE™ electronic service tool (Version 7.6.2 or higher) to communicate with any engine equipped with the new CENSE™ CM2330 module. A TSB has been published to show how advanced ECM data can be collected via INSITE™. Trend logs, Engine Protection logs, Enhanced Fault Snapshot logs, and Start-Stop logs should be downloaded every time a work order is collected. Reference TSB130149 for more information.

The Next Generation CENSE™ CM2330 module does **not** require the trend logs to be cleared out regularly. Once it has reached full capacity, the new module will start making space for new data by deleting the oldest data points in the memory.

There will be seventeen new fault codes for the Next Generation CENSE™ CM2330 product. They will be replacing existing CM530 fault codes. All customer equipment collecting fault code information from the CENSE™ module **must** update their list to reflect these changes. Reference TSB130148 for more information.

A new ROM boot cable for the Next Generation CENSE™ CM2330 module has also been released and is available to the field. The part number for this tool can be found in a summary table below.

Changing to the Next Generation CENSE™ CM2330 module will result in the loss of RS422 and RS232 interface support. Any original equipment manufacturer (OEM) or customer equipment utilizing these interfaces will need to upgrade to J1939-capable equipment.

As part of the migration to the new CENSE™ CM2330 module, a standard numbering system for exhaust gas temperature (EGT), intake manifold temperature (IMT), and intake manifold pressure (IMP) sensors has been adopted. Detailed old vs. new information is shown toward the end of this communication.

Although all engines that are currently in service will eventually be upfitted with the Next Generation CENSE™ CM2330 module, first engine serial number (ESN) information for QSK/QSV/KV engines equipped with the new module can be found at the end of this communication. All subsequent ESNs will be equipped with the Next Generation CENSE™ CM2330 from the plant.

Note : To avoid confusion in the field and false non-communication issues, Cummins Inc. recommends removing all RS232 connection points from the engine and/or truck cab as soon as the Next Generation CENSE™ CM2330 is installed.

Note : After upgrading to the Next Generation CENSE™ CM2330 module, field personnel might not be able to communicate to the new module from the cab if a J1939 port is not already installed in the cab. To support communication with the CENSE™ CM2330 module from the cab, be sure that a J1939 data link connection from the engine harness is properly extended. Contact a local application engineer for instructions to extend the connection.

Obsoleted Part Numbers						
Part Name			Part Number			
Engine Control Module			3098771			
K38 Engine Harness			3643114			
K38 Engine Harness			3643115			
K50 Engine Harness			3644538			
K50 Engine Harness			3644539			
New Part Numbers						
Part Name			Part Number			
Engine Control Module			4326926			
K38 Engine Harness			4328451			
K50 Engine Harness			4323226			
J1939 Upfit Jumper Harness			4328522			
9-Pin Service Connector L-Bracket			3649861			
CM2330 ROM Boot Cable			5298707			
Platform	Old ECM Code	New ECM Code	Old File Number	New File Number	FR Option	Note
NTC-400	G10002	N/A	4922482	N/A	FR01288	Obsoleted

Platform	Old ECM Code	New ECM Code	Old File Number	New File Number	FR Option	Note
QST30	G50043	EP80004	4998247	4328553	FR05134	
QST30	G50041	EP80008	3092440	4308662	FR05150	
QST30	G50034	EP80001	3092882	4328554	FR05151	
QST30		EP80005		4308655	FR05144	
QST30	G50031	EP80006	3092208	4328555	FR05227	
QST30	G50037	N/A	4068125	4328556	FR05232	Obsoleted
QST30	G50042	EP80003	4295355	4328557	FR05233	
K50	G60082	EP60024	3099310	4328558	FR06075	
K50	G60320	EP60051.0 0	3098393	4335183	FR06097	
K50	G60321	N/A	3098392		FR06097	Obsoleted
K38	G60322	EP60023	3098997	4328559	FR06101	Obsoleted
K38	G60083	EP60023	3099305		FR06101	
K38	G60342	EP60036	2873437	4335184	FR06142	
K50	G60001	EP60037	3098391	4335185	FR06145	
K50	G60323	EP60037	3098390		FR06145	Obsoleted
K50	G60349	EP60037	3099306		FR06145	Obsoleted
K50	G60242	EP60037	3099496		FR06145	Obsoleted
K50	G60338	EP60037	4971018		FR06145	Obsoleted
K50	G60014	EP60027	3099081	4328560	FR06147	Obsoleted
K50	G60300	EP60027	3099309		FR06147	
K50	G60334	EP60028	4968511	4328561	FR06202	
K50	G60324	EP60029	3098731	4328562	FR06215	Obsoleted
K50	G60087	EP60029	3099311		FR06215	
K38	G60088	EP60030	3099304	4328563	FR06233	
K50	G60004	EP60038	3098737	4335186	FR06237	Obsoleted
K50	G60333	EP60038	3099308		FR06237	
QSK45	G60304	EP60010	4065692	4328564	FR06268	
K50	G60305	EP60031	3098779	4328565	FR06276	Obsoleted
K50	G60301	EP60031	3099307		FR06276	
QSK45	G60307	EP60044	4065694	4357403	FR06289	
QSK45	G60329	EP60002	3636700	4328566	FR06289	
QSK45	G60308	EP60045	4065691	4357404	FR06291	
QSK45	G60309	EP60003	4080316	4328567	FR06291	
QSK78	G60327	EP60020	4080432	4328568	FR06302	
QSK78	G60330	EP60020	3637688		FR06302	Obsoleted

Platform	Old ECM Code	New ECM Code	Old File Number	New File Number	FR Option	Note
QSK45	G60310	EP60004	4065693	4328570	FR06305	
QSK60	G60098	EP60011	4016601	4328571	FR06325	
QSK60	G60225	EP60012	4016600	4328572	FR06327	
QSK60	G60099	EP60046	4080312	4357405	FR06329	
QSK60	G60100	EP60046	4080311		FR06329	Obsoleted
QSK60	G60101	EP60039	4080313	4335187	FR06329	
QSK60	G60129	EP60013	4086033	4328573	FR06330	
QSK60	G60102	EP60014	4086351	4328574	FR06331	Obsoleted
QSK60	G60178	EP60014			FR06331	
QSK60	G60254	EP60015	4080314	4328575	FR06339	Obsoleted
QSK60	G60255	EP60015	4080308		FR06339	Obsoleted
QSK60	G60256	EP60015	4065680		FR06339	
QSK60	G60257	EP60015	4080310		FR06339	Obsoleted
QSK60	G60233	EP60015	4100583		FR06339	Obsoleted
QSK60	G60296	EP60016	4080318	4328576	FR06340	Obsoleted
QSK60	G60297	EP60016	4080319		FR06340	Obsoleted
QSK60	G60298	EP60016	4065696		FR06340	
QSK60	G60260	EP60016			FR06340	Obsoleted
QSK60	G60325	EP60016	4101155		FR06340	Obsoleted
QSK60	G60245	EP60017	4065687	4328577	FR06359	
K38	G60339	EP60047	2866989	4357406	FR06367	
K38	G60350	EP60047	3637829		FR06367	Obsoleted
K50	G60340	EP60040	2866990	4335188	FR06367	
QSK60	G60289	EP60041	4065688	4335189	FR06378	
QSK60	G60292	EP60018	4101145	4328578	FR06378	Obsoleted
QSK60	G60331	EP60018	3637682		FR06378	
QSK60	G60312	EP60042	4080315	4335190	FR06379	
QSK60	G60314	EP60019	4065681	4328579	FR06379	
K50	G60299	EP60032	3099306	4328580	FR06389	
K50	G60337	EP60032	4971017		FR06389	Obsoleted
QSK45	G60140	EP60005	4085912	4328581	FR06404	
QSK45	G60142	EP60048	4085914	4357407	FR06406	
QSK45	G60143	EP60006	4085915	4328582	FR06406	
K50	G60345	EP60033	2896429	4328583	FR06426	
QSK45	G60336	EP60009	4101300	4328584	FR06443	

Platform	Old ECM Code	New ECM Code	Old File Number	New File Number	FR Option	Note
QSK60	G60227	EP60043	4016521	4335191	FR06458	
QSK60	G60130	EP60001.0 1	4065689	4323412	FR06554	
QSK45	G60302	EP60049	4065695	4357408	FR06640	
QSK60	G60123	EP60025	4085921	4328586	FR06687	
QSK78	G60328	EP60022	4101100	4328587	FR06704	
QSK45	G60341	EP60007	3635475	4328588	FR06707	
QSK60	G60335	EP60026	3636051	4328589	FR06714	
QSK78	G60344	EP60021	2886091	4328569	FR06725	
K38	G60332	EP60034	3637829	4328590	FR06819	
QSK45	G60311	EP60008	4065690	4328591	FR06979	
QST30		EP80007	4067824	4328592		
K50	G60348	EP60035	3099306	4335177	FR06356	
QSK60 - Gas		EP60050			FR6056	
QSK60 - Gas		EP60050			FR6587	
QSK60 - Gas		EP60050			FR6802	
QSK60 - Gas		EP60050			FR6803	
QSK60 - Gas		EP60050			FR6804	
QSK60 - Gas		EP60050			FR6805	
QSK60 - Gas		EP60050			FR6806	
QSK60 - Gas		EP60050			FR6911	
QSK60 - Gas		EP60050			FR6912	
QSV91 Phase 1		EP80010			FR7309	
QSV91 Phase 1		EP80010			FR7310	
QSV91 Phase 1		EP80010			FR7311	

Platform	Old ECM Code	New ECM Code	Old File Number	New File Number	FR Option	Note
QSV91 Phase 1		EP80010			FR7312	
QSV91 Phase 1		EP80010			FR7313	
QSV91 Phase 1		EP80010			FR7314	
QSV91 Phase 1		EP80010			FR7315	
QSV91 Phase 1		EP80010			FR7316	
QSV91 Phase 1		EP80010			FR7317	
QSV91 Phase 1		EP80010			FR7318	
QSV91 Phase 1		EP80010			FR7319	
QSV91 Phase 1		EP80010			FR7320	
QSV91 Phase 1		EP80010			FR7321	
QSV91 Phase 1		EP80010			FR7322	
QSV91 Phase 1		EP80010			FR7323	
QSV91 Phase 1		EP80010			FR7324	
QSV91 Phase 1		EP80010			FR7325	
QSV91 Phase 1		EP80010			FR7385	
QSV91 Phase 2		EP80009			FR07276	
QSV91 Phase 2		EP80009			FR07281	
QSV91 Phase 2		EP80009			FR07296	
QSV91 Phase 2		EP80009			FR07297	

Platform	Old ECM Code	New ECM Code	Old File Number	New File Number	FR Option	Note
QSV91 Phase 2		EP80009			FR07303	
QSV91 Phase 2		EP80009			FR07306	
QSV91 Phase 2		EP80009			FR07328	
QSV91 Phase 2		EP80009			FR07329	
QSV91 Phase 2		EP80009			FR07332	
QSV91 Phase 2		EP80009			FR07335	
QSV91 Phase 2		EP80009			FR07336	
QSV91 Phase 2		EP80009			FR07339	
QSV91 Phase 2		EP80009			FR07340	
QSV91 Phase 2		EP80009			FR07343	
QSV91 Phase 2		EP80009			FR07344	
QSV91 Phase 2		EP80009			FR07347	
QSV91 Phase 2		EP80009			FR07348	
QSV91 Phase 2		EP80009			FR07349	
QSV91 Phase 2		EP80009			FR07350	
QSV91 Phase 2		EP80009			FR07351	
QSV91 Phase 2		EP80009			FR07352	
QSV91 Phase 2		EP80009			FR07355	
QSV91 Phase 2		EP80009			FR07356	

Platform	Old ECM Code	New ECM Code	Old File Number	New File Number	FR Option	Note
QSV91 Phase 2		EP80009			FR07359	
QSV91 Phase 2		EP80009			FR07360	
QSV91 Phase 2		EP80009			FR07361	
QSV91 Phase 2		EP80009			FR07362	
QSV91 Phase 2		EP80009			FR07377	
QSV91 Phase 2		EP80009			FR07378	
QSK38 - Gas		EP60052			FR6691	
QSV91 TVAB		EP80011			FR7257	

Old CENSE™ Numbering for Exhaust Gas Temperature (EGT) Sensors	New CENSE™ CM2330 Numbering for Exhaust Gas Temperature (EGT) Sensors
Cylinder Number 1 Left Bank	Cylinder Number 1 (A1)
Cylinder Number 1 Right Bank	Cylinder Number 2 (B1)
Cylinder Number 2 Left Bank	Cylinder Number 3 (A2)
Cylinder Number 2 Right Bank	Cylinder Number 4 (B2)
Cylinder Number 3 Left Bank	Cylinder Number 5 (A3)
Cylinder Number 3 Right Bank	Cylinder Number 6 (B3)
Cylinder Number 4 Left Bank	Cylinder Number 7 (A4)
Cylinder Number 4 Right Bank	Cylinder Number 8 (B4)
Cylinder Number 5 Left Bank	Cylinder Number 9 (A5)
Cylinder Number 5 Right Bank	Cylinder Number 10 (B5)
Cylinder Number 6 Left Bank	Cylinder Number 11 (A6)
Cylinder Number 6 Right Bank	Cylinder Number 12 (B6)
Cylinder Number 7 Left Bank	Cylinder Number 13 (A7)
Cylinder Number 7 Right Bank	Cylinder Number 14 (B7)
Cylinder Number 8 Left Bank	Cylinder Number 15 (A8)
Cylinder Number 8 Right Bank	Cylinder Number 16 (B8)
Cylinder Number 9 Left Bank	Cylinder Number 17 (A9)
Cylinder Number 9 Right Bank	Cylinder Number 18 (B9)

K38, K50 Engines Only		
Old CENSE™ CM530 Numbering for Intake Manifold Temperature (IMT) Sensors	New CENSE™ CM2330 Numbering for Intake Manifold Temperature (IMT) Sensors	Fault Code Associated in New CM2330
Left Bank Front Intake Manifold Temperature	Intake Manifold Air Temperature #1	154
Right Bank Front Intake Manifold Temperature	Intake Manifold Air Temperature #3	161
Left Bank Rear Intake Manifold Temperature	Intake Manifold Air Temperature #2	157
Right Bank Rear Intake Manifold Temperature	Intake Manifold Air Temperature #4	164
QSK45, QSK60 Engines Only		
Old CENSE™ CM530 Numbering for Intake Manifold Temperature (IMT) Sensors	New CENSE™ CM2330 Numbering for Intake Manifold Temperature (IMT) Sensors	Fault Code Associated in New CM2330
Left Bank Front Intake Manifold Temperature	Intake Manifold Air Temperature #1	Remote (CM500)
Right Bank Front Intake Manifold Temperature	Intake Manifold Air Temperature #3	161
Left Bank Rear Intake Manifold Temperature	Intake Manifold Air Temperature #2	157
Right Bank Rear Intake Manifold Temperature	Intake Manifold Air Temperature #4	164
QSK78 Engines Only		
Old CENSE™ CM530 Numbering for Intake Manifold Temperature (IMT) Sensors	New CENSE™ CM2330 Numbering for Intake Manifold Temperature (IMT) Sensors	Fault Code Associated in New CM2330
Left Bank Front Intake Manifold Temperature	Intake Manifold Air Temperature #1	Remote (CM500)
Right Bank Front Intake Manifold Temperature	Intake Manifold Air Temperature #3	161
Left Bank Middle Intake Manifold Temperature	Intake Manifold Air Temperature #5	2243
Right Bank Middle Intake Manifold Temperature	Intake Manifold Air Temperature #6	2247
Left Bank Rear Intake Manifold Temperature	Intake Manifold Air Temperature #2	157

QSK78 Engines Only		
Old CENSE™ CM530 Numbering for Intake Manifold Temperature (IMT) Sensors	New CENSE™ CM2330 Numbering for Intake Manifold Temperature (IMT) Sensors	Fault Code Associated in New CM2330
Right Bank Rear Intake Manifold Temperature	Intake Manifold Air Temperature #4	164

Old CENSE™ Numbering for Intake Manifold Pressure (IMP) Sensors	New CENSE™ CM2330 Numbering for Intake Manifold Pressure (IMP) Sensors
Left Bank Intake Manifold Pressure	Intake Manifold Pressure #1
Right Bank Intake Manifold Pressure	Intake Manifold Pressure #2
Platform	ESN First
QSK78 HPI	66302969
QSK60 HPI	33199842
K50	33199574
QSK60 - Gas	33199692
QSV91 - Gas	66302996

Document History

Date	Details
2013-9-25	Module Created
2014-3-11	none
2018-7-11	Added additional CENSE™ tables.
2018-10-16	Corrected pg10 table for QSK78 Engines

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